3. Let ΔABC be equilateral. If D is a point on BC and AD is the internal bisector of $\angle A$. Using Angle Bisector Theorem, $\frac{BD}{DC}$ is _____.
4. The _____ of an angle of a triangle divides the opposite side internally in the ratio of the corresponding sides containing the angle.
5. If the median AD to the side BC of a ΔABC is also an angle bisector of $\angle A$ then $\frac{AB}{AC}$ is _____.

4.3.2 Construction of triangle

We have already learnt in previous class how to construct triangles when sides and angles are given.

In this section, let us construct a triangle when the following are given :

(i) the base, vertical angle and the median on the base

(ii) the base, vertical angle and the altitude on the base

(iii) the base, vertical angle and the point on the base where the bisector of the vertical angle meets the base.

First, we consider the following construction,

Construction of a segment of a circle on a given line segment containing an angle θ

Construction

Step 1: Draw a line segment $\overline{AB}$.

Step 2: At A, take $\angle BAE = \theta$ Draw AE.

Step 3: Draw, $AF \perp AE$.

Step 4: Draw the perpendicular bisector of AB meeting AF at O.

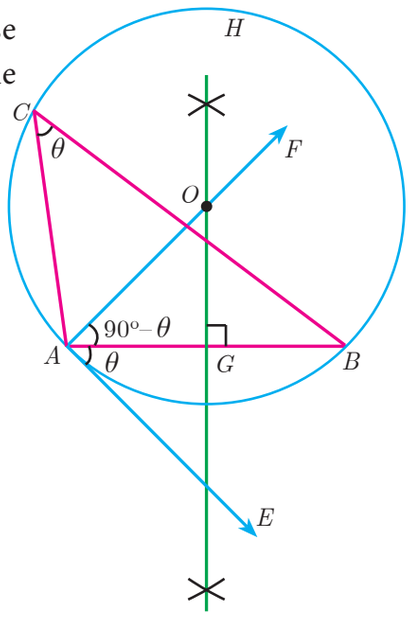


Fig. 4.41

Step 5: With O as centre and OA as radius draw a circle ABH.

Step 6: Take any point C on the circle, By the alternate segments theorem, the major arc ACB is the required segment of the circle containing the angle θ.

Note

If $C_1, C_2, \ldots$ are points on the circle, then all the triangles ΔBAC_1, $\Delta BAC_2, \ldots$ are with same base and the same vertical angle.

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